YIYOU SUN

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Education

University of Wisconsin-Madison Sept. 2018 - July. 2023
PhD. in Computer Science, Minor in Math *Madison, US*

- GPA 3.85/4.0. Advised by Prof. Sharon (Yixuan) Li.

Harbin Institute of Technology Sept. 2014 - Jun. 2018
B.S. in Computer Science *Harbin, China*

- GPA 93.67/100. Rank **1st** among 97 students.

Ph.D. Research Overview

– *We aim to pave the way to a reliable Open-world Machine Learning system.*

Out-of-distribution (OOD) Detection: In an open-world, training and testing distribution for deep neural networks can be different. We want to build robust machine learning system that can identify OOD samples in inference time.

Open-world Representation Learning: As an additional step for detecting OOD, we want the model to learn OOD samples from the novel classes. Under the setting, the model should ideally learn distinguishable representations—not only for the known classes but also the novel categories.

Interpretability: Build an interpretable machine learning system, particularly in visualizing and quantitating the model's inference mechanism or constructing one inherently sparse, simulatable, modular, and thus, interpretable.

Academic Experience

University of Wisconsin-Madison Sept 2020 - July 2023
Research Assisstant *Madison, WI*

- Advised by Prof. Sharon (Yixuan) Li.
- Finished thesis "*Detecting and Learning Out-of-Distribution Data in the Open-world: Algorithm and Theory*", investigating two essential steps for open-world machine learning: Out-of-distribution (OOD) Detection and Open-world Representation Learning (ORL).

University of Wisconsin-Madison Sept 2018 - May 2020
Research Assisstant *Madison, WI*

- Advised by Prof. Vikas Singh.

MIT: Computer Science and Artificial Intelligence Laboratory Dec 2017 - May 2018
Undergraduate Research Assisstant *Cambridge, MA*

- Supervised by Dr. Bolei Zhou (now at UCLA).

Harvard University: DASlab July 2017 - May 2018
Undergraduate Research Assisstant *Cambridge, MA*

- Supervised by Prof. Stratos Idreos.

Industrial Experience

NEC Laboratories America Inc. Aug. 2023 - Current
Researcher *Plainsboro, NJ*

- Report to Dr. Haifeng Chen.

Google Research May 2022 - Sept 2022
Research Intern *Mountain View, CA*

- Supervised by Dr. Yaojie Liu and Dr. Xiaoming Liu.

Facebook Integrity Team June 2021 - Sept 2021
Research Intern *Remote*

- Supervised by Dr. Sethuraman Sankaran and Dr. Ser-Nam Lim.

Facebook AI June 2020 - Sept 2020
Research Intern *Remote*

- Supervised by Dr. Harry Yang and Dr. Ser-Nam Lim.

First/co-first Authored Publications

[1] (**Spotlight**) **Y. Sun**, S. Zhenmei, Y. Li. "*A Graph-Theoretic Framework for Understanding Open-World Semi-Supervised Learning*". In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS), 2023.

[2] **Y. Sun**, S. Zhenmei, L. Yingyu, Y. Li. "*When and How Does Known Class Help Discover Unknown Ones? Provable Understanding Through Spectral Analysis*". In Proceedings of the International Conference on Machine Learning (ICML), 2023.

[3] **Y. Sun**, Y. Liu, X. Zhu, Y. Li, W. Chu. "*Rethinking Domain Generalization for Face Anti-spoofing: Separability and Alignment*". In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.

[4] **Y. Sun**, Y. Ming, X. Zhu, Y. Li. "*Out-of-distribution Detection with Deep Nearest Neighbors*". In Proceedings of the International Conference on Machine Learning (ICML), 2022.

[5] **Y. Sun**, C. Guo, Y. Li. "*ReAct: Out-of-distribution Detection with Rectified Activations.*" In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS), 2021.

[6] **Y. Sun**, Y. Li. "*Dice: Leveraging sparsification for out-of-distribution detection.*" In Proceedings of the the European Conference on Computer Vision (ECCV), 2022.

[7] **Y. Sun**, Y. Li. "*OpenCon: Open-world Contrastive Learning with Wild Unlabeled Data*". Transactions on Machine Learning Research (TMLR), 2022.

[8] **Y. Sun**, S. Ravi, V. Singh. "*Adaptive Activation Thresholding: Dynamic Routing Type Behavior for Interpretability in Convolutional Neural Networks.*" In Proceedings of the IEEE Conference on Computer Vision (ICCV), 2019.

[9] B. Zhou*, **Y. Sun***, D. Bau*, A. Torralba. "*Interpretable Basis Decomposition for Visual Explanation.*" In Proceedings of the European Conference on Computer Vision (ECCV), 2018. (* indicates equal contribution)

Collaborative Publications

[10] X. Du, **Y. Sun**, X. Zhu, Y. Li "*Dream the Impossible: Outlier Imagination with Diffusion Models.*" In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS), 2023.

[11] Z., Jingyang, J. Yang, P. Wang, H. Wang, Y. Lin, H. Zhang, **Y. Sun**, X. Du, K. Zhou, W. Zhang, Y. Li, Z. Liu, Y. Chen, H. Li. "OpenOOD v1.5: Enhanced Benchmark for Out-of-Distribution Detection". arXiv preprint arXiv:2306.09301, 2023.

[12] Y. Ming, **Y. Sun**, O. Dia, and Y. Li. "*How to Exploit Hyperspherical Embeddings for Out-of-Distribution Detection?.*" International Conference on Learning Representations (ICLR), 2023.

[13] Y. Ming, Z. Cai, J. Gu, **Y. Sun**, W. Li, Y. Li. "*Delving into OOD Detection with Vision-Language Representations.*" In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS), 2022.

[14] J., Yang, P. Wang, D. Zou, Z. Zhou, K. Ding, W. Peng, H. Wang, G. Chen, B. Li, **Y. Sun**, X. Du, K. Zhou, W. Zhang, D. Hendrycks, Y. Li, Z. Liu. "*OpenOOD: Benchmarking Generalized Out-of-Distribution Detection.*" In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS), 2022.

[15] B. Zhou, **Y. Sun**, D. Bau, A. Torralba. "*Revisiting the Importance of Individual Units in CNNs via Ablation*". arXiv preprint arXiv:1806.02891, 2018.

Other Projects

[16] **Y. Sun**, B. Joseph, A. Deatsh, R. Jeraj, Y. Li. "*LOOD: Localization-based Uncerntainty Estimation for Medical Imaging.*" ICML Workshop on Distribution-Free Uncertainty Quantification (ICML DFUQ), 2021.

[17] **Y. Sun**, T. Su, Z. Tu. "*Faster R-CNN Based Autonomous Navigation for Vehicles in Warehouse*". In 2017 IEEE International Conference on Advanced Intelligent Mechatronics (AIM), pages 1639-1644, July 2017.

[18] S. Idreos, K. Zoumpatianos, M. Athanassoulis, N. Dayan, B. Hentschel, M. S. Kester, D. Guo, L. M. Maas, W. Qin, A. Wasay, and **Y. Sun**. "*The Periodic Table of Data Structures*". IEEE Data Eng. Bull., 41(3):64-75, 2018.

[19] S. Idreos, K. Zoumpatianos, S. Chatterjee, W. Qin, A. Wasay, B. Hentschel, M. Kester, N. Dayan, D. Guo, M. Kang, **Y. Sun**. "*Learning Data Structure Alchemy*". Bull. of the IEEE Computer Society Technical Committee on Data Engineering. 42(2), 2019.

Community Services

Journal Reviewer

- Review for Pattern Recognition in 2021, 2022.
- Review for IEEE Transactions on Cybernetics in 2022.
- Review for IEEE Transactions on Multimedia in 2023.
- Review for International Journal of Computer Vision (IJCV) in 2022, 2023.
- Review for IEEE Transactions on Image Processing (TIP) in 2023.

Conference Reviewer

- Review for Conference on Computer Vision and Pattern Recognition (CVPR) in 2022, 2023.
- Review for International Conference on Machine Learning (ICML) in 2022, 2023.
- Review for Conference on Neural Information Processing Systems (NeurIPS) in 2022 (**Top Reviewer Award**), 2023.
- Review for European Conference on Computer Vision (ECCV) in 2022.
- Review for International Conference on Learning Representations (ICLR) in 2022, 2023.
- Review for Association for the Advancement of Artificial Intelligence (AAAI) in 2022, 2023.

Talks

The Theoretical Foundation for the Open-world Representation Learning: A Spectral View Oct 2023
Invited Talk @ UIUC ML Seminar *Remote*

Out-of-distribution (OOD) Generalization and Detection in AI Risk Management Oct 2023
Invited Talk @ Beijing Academy of Artificial Intelligence *Remote*

Detecting and Learning Out-of-Distribution Data in the Open world July 2023
Ph.D. Oral Defense *Madison, WI*

Rethinking Domain Generalization for Face Anti-spoofing: Separability and Alignment Sept 2022
Intern Presentation @ Google Research *Mountain View, CA*

Introduction on Interpretable Deep Learning Oct 2020
Guest lecture @ CS 839 – Special Topics in AI: Deep Learning *Madison, WI*